nanoker

CERAMIC TECHNICAL DATA SHEET

# KSIALON

## Silicon Nitride-based Ceramic

### KEY PROPERTIES

- High strength
- High toughness
- High temperature stability
- Very high thermal shock resistance
- Corrosion resistance to non-ferrous metals

### APPLICATIONS

- Structural parts for machinery
- Welding pins
- Welding rolls
- Gas nozzles
- Riser tubes (foundries)

### MAIN PROPERTIES

| Property | Value | Unit |
|---|---|---|
| Density | 3.2 | g/cm³ |
| Flexural Strength | 900 | MPa |
| Fracture Toughness | 7.0 | MPa·√m |
| Young Modulus | 280 | GPa |
| Thermal Conductivity | 25 | W/m·K |
| CTE (20–1000ºC) | 3.5 | $x10^{-6}$ $K^{-1}$ |
| Electrical Resistivity | $>10^{12}$ | Ω·cm |
| Weibull Modulus | 15 | - |

** All properties measured at 20ºC unless otherwise stated.*
*** Measured in a wall thickness of 2 mm*

### OPERATING CONDITIONS

| | | |
|---|---|---|
| Max working temp (air) | 1100 | ºC |
| Recomended temp | <1000 | ºC |
| Continuous temp | 950 | ºC |

### ELECTRICAL BEHAVIOR

Insulator

### MATERIAL INFORMATION

| | |
|---|---|
| Material type | Silicon Nitrided-based (Sialon) |
| Structure | Dense ceramic |
| Key feature | Thermal shock resistance |
| Machinability | Low |
| Color | Dark grey |

Local overheating may occur at electrical contact points due to current concentration and geometry effects.



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